Geophys. J. R. astr. Soc. (1972) 27, 517–545.

A Palaeomagnetic Study of the Mull Regional Dyke Swarm

J. M. Ade-Hall*, P. Dagley, R. L. Wilson, A. Evans, A. Riding, P. J. Smith, R. Skelhorne and T. Sloan

(Received 1971 September 14)†

Summary

A detailed palaeomagnetic investigation has been made of the Mull (Scotland) regional dyke swarm. 165 dykes were sampled in a 7-km section across the swarm from Quinish to Ardmore in northern Mull. A major question arising from the results of the investigation is whether the cleaned NRM directions for the dykes are equivalent to the original cooling TRM directions. The question arises since evidence from several lines shows that the dykes have experienced considerable hydrothermal alteration, with reheating temperatures averaging 200°C and major changes in the iron titanium oxides responsible for the NRM. Surprisingly, the answer appears to be that original cooling TRM directions are probably still well represented by cleaned NRM directions.

Notwithstanding the large number of palaeomagnetic directions obtained from the dykes, no simple picture of the Palaeogene geomagnetic field emerges. Thus, continental drift cannot be estimated from the data. The explanation for the distribution of directions appears to rest between either a short time of activity for the whole British Tertiary igneous province with non-representative sampling by the activity in individual areas, or the existence of relatively long periods of non-centred axial dipole behaviour for the geomagnetic field during the Palaeogene.

1. Introduction

One uncertainty concerning the basic assumption of palaeomagnetic interpretation dominates this study, that the geomagnetic field direction at the time of initial cooling of a small volcanic unit can be isolated, and much of the paper will be oriented towards answering this doubt. The question that must be asked is whether a series of dykes, probably initially characterized by Curie points largely in the 100–200°C temperature interval, can after possibly later reheating to 200°C be made to yield information about the geomagnetic field at the time of initial cooling. If this doubt can be resolved satisfactorily, then consideration can be given to the pattern of geomagnetic field behaviour at the time of the intrusion of the dykes. Once the question has been resolved for this particular dyke swarm, from Mull, Scotland, then palaeomagnetic results from other less highly reheated parts of the British Tertiary

* Now at the Department of Geology, Dalhousie University, Halifax, Nova Scotia, Canada.

† Received in original form 1970 September 21.

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volcanic province can be taken as representing time of cooling geomagnetic field directions, and a picture can be built up of the field behaviour over an area of some 2×10^5km. for part of the Palaeogene.

2. The dyke swarm

In this section we summarize the conventional geology of the dyke swarm and elaborate on the post consolidation alteration history as this is particularly relevant to the present magnetic state of the dykes.

2.1 *General Geological Setting*

The dyke swarm studied is the classic NW–SE trending regional swarm of Mull, Scotland (Bailey *et al*, 1924; Richey 1939). Dykes belonging to the swarm extend at least from the Outer Hebrides to North-east England, a distance of about 550 km. A focal point along the line of the swarm is the Mull central intrusive complex, which is thought to mark the roots of a large Palaeogene volcano (Judd 1889; Bailey *et al.*, 1924; Walker 1970). Near this centre the dyke swarm is at its narrowest and contains the maximum recorded densities of dykes, with equivalent crustal extension of up to 10 per cent (Skelhorn, private communication). The swarm was sampled along the north-west coast of Mull between Quinish (56° 36·0′ N, 353° 45·5′ E) and Ardmore (56° 39·0′ N, 353° 51·0′ E). The traverse from which dykes were sampled probably contains some two thirds of the total dykes comprising the regional swarm in the area and includes the belts of maximum density of intrusion. In all a total of 165 dykes was sampled, both olivine dolerites and tholeiites being well represented. Dykes are near to vertically oriented; a small but systematic hade variation has been recognized by Sloan (1970). Average dyke trend is N330 E and average thickness 1·58 m with a range from 0·10 to 9·10 m. The country rock is sub-horizontal Tertiary plateau lavas, with the exception of rare cases when an older dyke forms the country rock for a younger dyke. There seems to be little doubt that the age of the British Tertiary volcanic province is Palaeogene, but within this range the ages and lifetimes of many individual features, such as the Mull dyke swarm, are still unknown. Geological evidence shows that basic dykes were intruded throughout the life of the Mull central intrusive complex, but the absolute duration of activity here is as yet undetermined.

Two reasons led to the choice of the Quinish–Ardmore section for palaeomagnetic sampling. In the first place it was well separated from previously recognized areas of high alteration surrounding the central intrusive complex. Secondly, almost continuous exposure in a section approximately normal to the trend of the dyke swarm was ensured. In fact, exposure was lacking only at the head of Loch Mingary (56° 37·3′ *N*, 353° 47·0′ E) for about 200 m, or about 2·5 per cent of the total traverse length of about 7·1 km.

2.2 *Post consolidation alteration of the dykes*

On consolidation, an opaque mineral facies responsible for the magnetic properties of the dykes of rather simple type and remarkably widespread distribution was formed. The facies consisted of dominant titanomagnetite and minor iron sulphides and ilmenite. The general occurrence of small amounts of iron sulphides and the uniformly low separate ilmenite content contrasts strongly with the immediately post-consolidation opaque mineralogy of many sub-aerial basalt lavas (Watkins & Haggerty 1968; Ade-Hall & Lawley 1970).

From the results of Ade-Hall (1964), Carmichael & Nicholls (1967) Smith (1967a) Creer & Peterson (1969) and others where a remarkably widespread limited compositional range is found to occur for basaltic titanomagnetites, we can suppose a similar immediately post-consolidation composition for the typical dyke titanomagnetite.

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The mean composition of the rather tight clustering of results is near to 0·3 $Fe_3O_4.0.7Fe_2TiO_4$, that is, closer to the ulvospinel end of the magnetite-ulvospinel solid solution series. Calculation of equivalent Curie points for these titanium rich titanomagnetites, on the assumption of stoichiometry, yields a limited range of values peaked in the 100–200°C interval. Note well this low interval for the theoretical Curie points of the immediately post-consolidation titanomagnetite as it forms an important point for the arguments that follow concerning the history of the natural magnetism of the dykes.

2.2.1 *Deuteric oxidation of the dykes.* The first type of alteration to basaltic rocks following consolidation is the well-known deuteric oxidation process. Simply, this involves the lava rock burning in its own or an acquired gas content. Sato & Wright (1966) have followed this process as it happens in cooling contemporary Hawaiian lava lakes—in some parts of the uppermost parts of these lava lakes oxygen fugacity rises dramatically over the 800–500 °C cooling interval; this being accompanied by reddening of the rock. Changes in the magnetic titanomagnetite during this process have been traced by Grommé, Wright & Peck (1969). The first stage of titanomagnetite deuteric oxidation results in the sub-solidus exsolution of ilmenite lamellae trellises, leaving near pure magnetite as the host. Further oxidation results in the splitting of the ilmenite lamellae into titanohematite and ferri-rutile while finally a pseudomorph consisting of pseudobrookite, titanohematite and rutile completely replaces the original titanomagnetite. The titanomagnetite deuteric oxidation range has been classified into six stages by Wilson, Haggerty & Watkins (1968) and a mean magnetite deuteric oxidation number, M, has been defined. M runs from 1·00 (deuteric oxidation absent) to 6·00 (complete replacement of titanomagnetite by pseudobrookite, titanohematite and rutile). The striking feature of the deuteric oxidation of the Mull dykes, in contrast to subaerial basalts, is their uniformly low degree of deuteric oxidation (Ade-Hall & Lawley, 1970). This low degree of deuteric oxidation, coupled with low separate ilmenite content and nearly ubiquitous iron sulphides bears witness to the low oxygen fugacity of the magma gas phase during initial crystallization and cooling down to at least 500 °C. In fact, of 187 specimens examined from 57 dykes, no specimen had a magnetite deuteric oxidation number in excess of 2·00, (corresponding to the mild state of oxidation where grains have only occasional ilmenite lamellae). In 141 of the 187 specimens (75·4 per cent), M is unity, i.e. visible signs of deuteric oxidation are completely absent.

Now, one of the magnetic effects of the deuteric oxidation of titanomagnetite is the development, of a high (500–600°C) Curie point phase by oxidation of a fraction of the magnetite grains, or, in some cases general Curie point rise (Ade-Hall, Wilson & Smith 1965; Babkine 1965; Lawley & Ade-Hall 1971). The widespread *absence* of deuteric oxidation in the dykes means that their initially low (100°–200°C) Curie points missed this early opportunity for oxidation-induced elevation and the initial cooling TRM of the rocks remained apparently vulnerable to later moderate reheating.

2.2.2 *Regional hydrothermal alteration of the dykes.* The second stage in the alteration of the dykes and their titanomagnetites took place in zeolite facies metamorphic conditions. These conditions are bounded by maximum temperatures of about 300°C and pressures of up to a few kilobars. The alteration process for the rock as a whole is conveniently described as regional hydrothermal alteration (Ade-Hall *et al.* 1971). Regional hydrothermal alteration may take place at a considerable time after unit consolidation, up to several million years in the case of lavas but possibly less for dykes: At a deep level in the near-vertical sheet comprising a dyke, hydrothermal alteration might follow immediately after deuteric oxidation—the cause of the alteration being chemical attack by hot ground water percolating into the dyke from the lava country rock. On the other hand dykes may, after intrusion, attain a

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deeper burial level, for example, by continued accumulation of the lava pile. Reheating temperatures in regional hydrothermal alteration conditions can surpass the 200°C peak temperature below which most unoxidized titanomagnetite initial Curie points occur and our first problem is to assess the degree of regional hydrothermal alteration experienced by the dyke swarm in our area.

Elsewhere (Walker 1960a, 1960b; Stoiber & Davidson 1959) the degree of regional hydrothermal alteration of basaltic lavas has been judged on a secondary mineral facies. A zonal scheme has been devised (Fig. 1) (Ade-Hall *et al.* 1971) where zones extend from zone A, describing largely secondary mineral free basalts that have experienced only ambient temperatures, to zone I, describing epidote bearing basalts that have experienced temperatures probably in excess of 300°C.

No secondary mineral studies have been made on the dykes in the Quinish–Ardmore area, although information is available for the lava country rock (Walker 1970). To assign the dykes to a regional hydrothermal alteration zone we use two related features, the state of alteration of the titanomagnetites, and the Curie point distribution, Ade-Hall *et al.* (1970, 1971) have shown that both a systematic progression of opaque mineralogical changes and a steady rise in Curie point accompanies increasing degree of regional hydrothermal alteration. The significant features of the dyke opaque mineralogy are the widespread occurrence of titanomagnetite

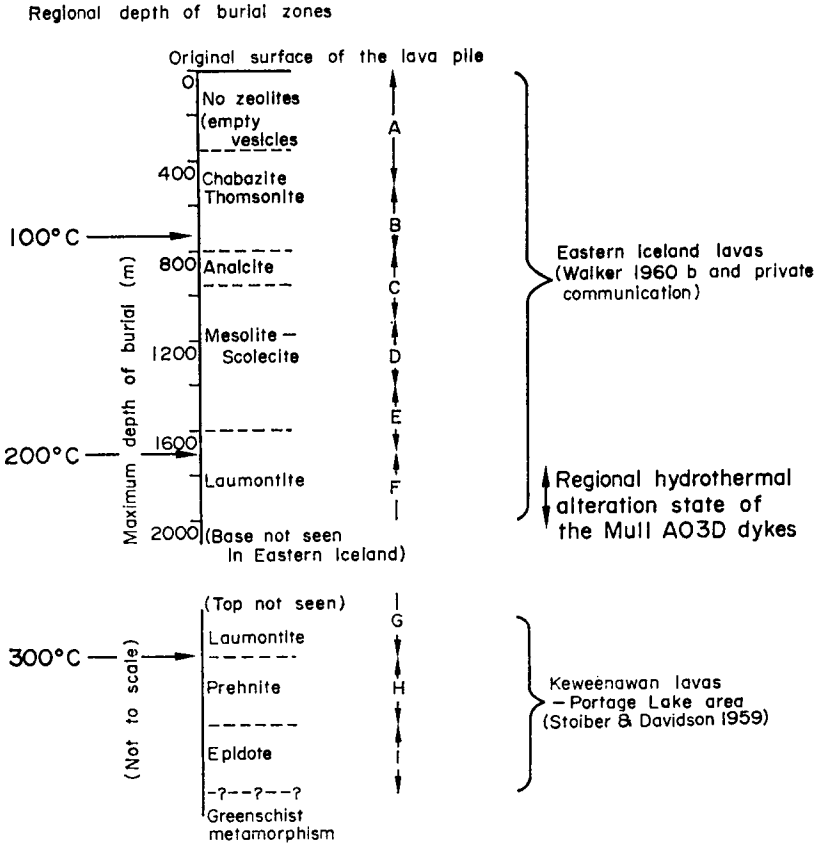


FIG. 1. The zonal scheme for the regional hydrothermal alteration of basaltic rocks, based on lava successions in Eastern Iceland and the Lake Superior region. Reheating temperatures and the position in the scheme of the Mull Regional dyke swarm in the Quinish–Ardmore section (code A03D) are indicated.

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granulation (in 29 (76 per cent) of 38 dykes examined for this feature) and the *unaltered* state of the rather rare ilmenite exsolution lamellae in titanomagnetite. The first feature, consisting of patches of fine ferri-rutile granules in titanomagnetite (Ade-Hall *et al.* 1968) first appears in D zone lavas while alteration of ilmenite lamellae to sphene ($CaTiSiO_5$) begins in G zone lavas. Thus the state of the dyke titanomagnetites suggests regional hydrothermal alteration within the zone D to zone G range, with a corresponding temperature interval of 150–250°C.

Using the Curie point distribution this zonal range can be narrowed somewhat. Fig. 2 shows distributions for the dykes and for Icelandic E zone and Mull G zone lavas. We see that the distribution for the dykes lies roughly midway between each of the lava distributions, and it seems reasonable to assign the dykes to zone F, with a corresponding reheating temperature of about 200°C. It is important to note that this is the maximum temperature of the estimated temperature interval for dyke post consolidation Curie points, because a reheating to above the Curie point has, in the past, been thought to destroy the original NRM.

Two other important questions regarding the regional hydrothermal alteration of the dykes need to be asked: Is there any lateral variation in degree of alteration and can the time of alteration be determined? The answer to the first query is that dyke titanomagnetite granulation and Curie points show rather irregular variations but that mean values for one-kilometre intervals across the swarm both show a gradual and rather irregular rise towards the north-eastern (Ardmore) end of the Section (Fig. 3). That is, average regional hydrothermal alteration was apparently a little higher at the Ardmore end of the section than at the Quinish end with respect to the present erosional level.

The second suggestion cannot be answered with certainty. Walker's study shows the country rock to be in the laumontite zone while less altered scolecite bearing lavas occur in the adjacent Calgary and Tobermory areas (Fig. 4). Thus the country rock lavas may also be in the zone F state of alteration of the dykes. This result is consistent with either of two possibililities: (i) The dykes were intruded into lavas already experiencing F zone conditions and were essentially immediately altered to the same degree as the lavas, or (ii) The dykes were intruded into lavas experiencing less severe than zone F conditions and both lavas and dykes were altered together at some later time in zone F conditions. The delay between dyke intrusion and maximum degree of alteration may well have been long enough for alteration to have taken place in a quite differently directed magnetic field from that responsible for the original TRM.

While there seems to be no clear way of choosing between these two possibilities there is a statistical argument in favour of the second one. It is unlikely that all the dykes were intruded during the maximum burial stage in the history of the lava pile. Those intruded earlier than the maximum burial stage will have been further altered at some time after intrusion. However, there is no way at present of separating early altered from later altered dykes.

We must also note the implication of the Curie point distribution; regional hydrothermal alteration of basaltic titanomagnetite is not simply responsible for new thermoremanent magnetization, but produces an altered magnetite with a higher Curie point. The remanent magnetization of this altered magnetite must depend both on the relative values of reheating temperatures and rising Curie point (Wilson & Smith 1968; Marshall & Cox 1971) and on the intrinsic magnetic properties of the altered magnetite. Regarding the intrinsic properties of the changing magnetic phase, it is important to bear in mind that it may well be a cation deficient form of titanomagnetite (Ade-Hall, Hubbard & Palmer 1970; Readman & O'Reilly 1971). There is some evidence that very cation deficient high titanium magnetites may in special circumstances show the property of self reversal of remanent magnetization (O'Reilly & Banerjee 1966; Ozima & Ozima 1967). However, the studies of Wilson & Smith

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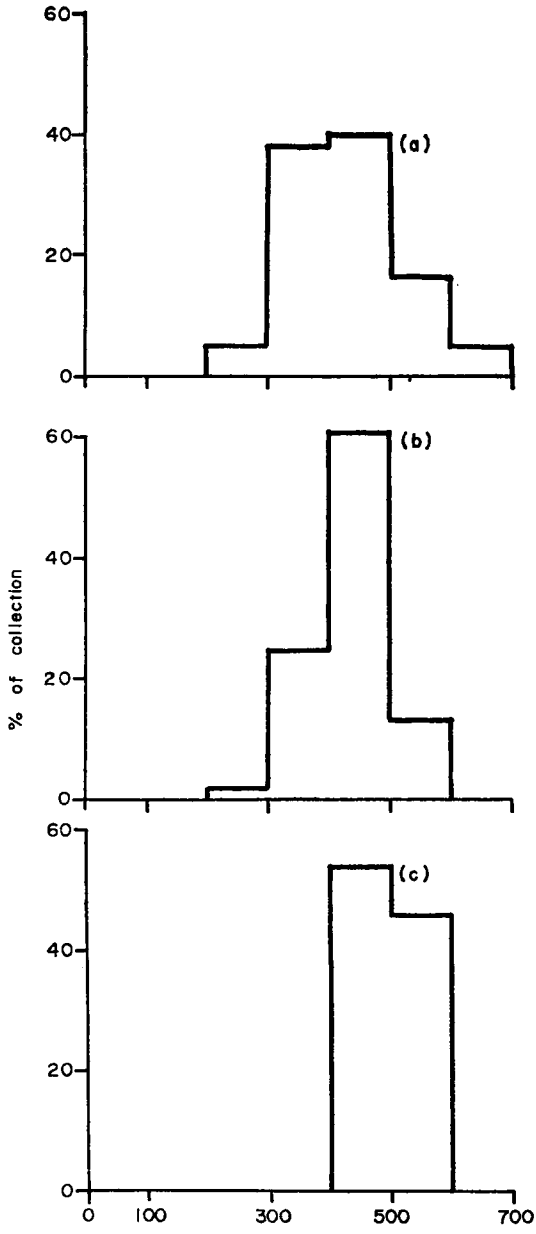


FIG. 2. (a) A comparison of the Curie point distributions for E regional hydrothermal alteration zone lavas for Eastern Iceland; (b) the Mull regional dyke swarm in the Qunish–Ardmore section; (c) G zone lavas from Mull.

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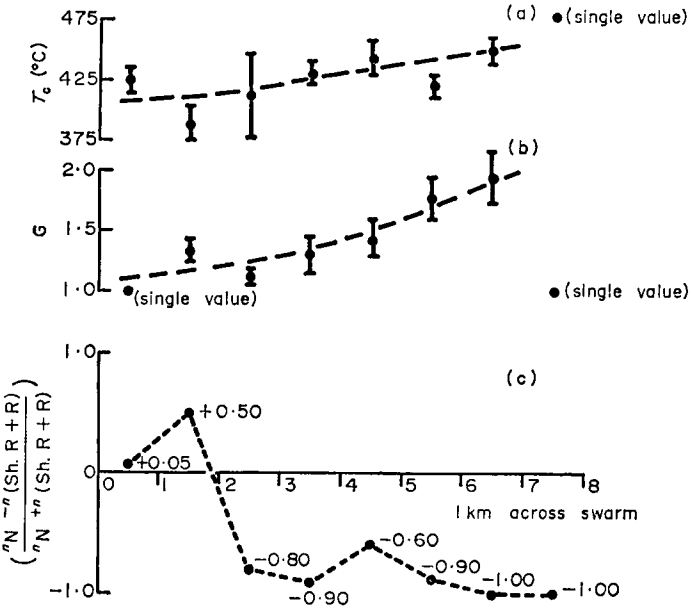


FIG. 3. Mean values for selected magnetic and opaque petrological properties for kilometre intervals across the strike of the dyke swarm. (a) Strong field Curie point; (b) Titanomagnetite granulation; (c) The polarity ratio $[n_N - n_{Sh.\,R+R}/n_N + n_{Sh.\,R+R})$ where n is the number of dykes in different polarity groups, i.e. difference between numbers of normal and reverse dykes as a fraction of the total number of dykes.

(1968) and Marshall & Cox (1971) indicate that, at least in laboratory conditions, NRM directions may be accurately preserved during the alteration of the magnetite.

3. Sampling

Specimens in the form of drill cores at least 5 cm long were obtained by means of portable diamond drills based on the design of the United States Geological Survey (Doell & Cox 1967). Cores of 2·5-cm diameter were drilled in fresh rock surfaces, frequently between high and low tide marks. Usually either four geographically oriented cores were taken from a dyke, or three cores from the dyke together with a baked specimen from the country rock directly adjacent to the dyke. Dyke locations were marked on U.K. Ordnance Survey, six inches to one mile scale (1 : 10 560) maps; County Series, Argyllshire (Isle of Mull) sheets 38 and 38A. Field notes include drill core orientation, dyke thicknesses and dyke and baked rock specimen positions relative to the nearest contact. Specimen numbers are of the form AO3D288-2 where the code AO3D describes the Mull dykes collection in the series of Liverpool palaeomagnetic collections, 288 the dyke number and 2 the specimen number. Where possible specimens were numbered with respect to the nearest dyke-baked rock contact, with specimen 1 as close as possible to the contact and specimen 2 and 3 at approximately a quarter and a half-dyke width respectively from the contact. Baked rock code numbers differ slightly from dyke specimen code numbers: for example AO3D288A5 describes the baked rock sample nearest to the contact of dyke 288, while AO3D288B6 indicates the next furthest away baked rock, and so on.

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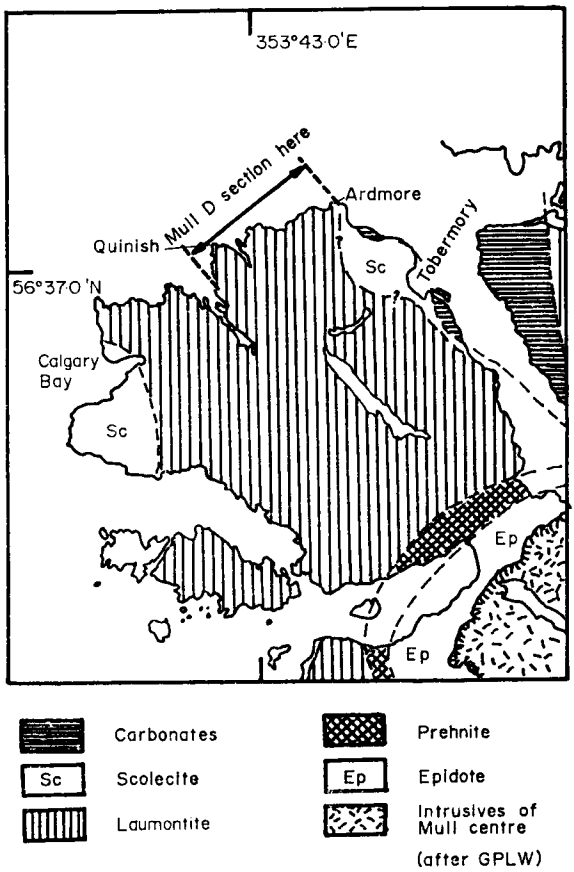


FIG. 4. The distribution of secondary minerals in the lavas of the Mull north-west part of the Isle of Mull and adjacent areas (reproduced from Walker 1970, with kind permission of the author).

4. Natural remanent magnetization measurements

Remanence measurements were made with an astatic magnetometer using 2·5 cm length cyclinders cut from the drill cores. The NRM of every dyke specimen was alternating field cleaned in 50 Oe steps to at least a peak field of 200 Oe. This value was chosen after tests with a trial group of specimens had shown that final directions were usually achieved by this field. Baked rock specimens, which were generally much harder magnetically than dyke specimens, were demagnetized in 100 Oe steps to at least a peak field of 400 Oe. Alternating field demagnetization was carried out to higher fields for specimens where NRM's were clearly still changing direction at the routine limiting fields. Dyke mean directions were obtained in the first instance by a standard computing procedure (Dagley & Ade-Hall 1970). Where computed α_{95} values exceeded 10° the data was re-examined and individual specimens or dykes deleted on the grounds that the data did not fit the physical requirement of within dyke internal consistency of directions. As a result of this re-examination the within dyke scatter of directions was considered excessive for 11 dykes and one or more aberrant specimens were removed from the averages for a further 25 dykes, leaving a minimum of two specimens per dyke in this group. In some cases the polarity of deleted dykes is clear, and has been recorded in Table 1, while α_{95} values for the dykes from which one or two specimens had been removed were with four exceptions,

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reduced to below 10°. Specimens removed from averages usually were either clearly different in direction of magnetization from the general direction shown by the other specimens, suggesting that an orientation or marking error had been made in the field, or were so soft magnetically that anhysteritic or other effects during demagnetization became excessive before a final direction was achieved. The directions for 154 of the original collection of 165 dykes are considered to be reliable.

Some of the dykes that have had directions modified or deleted featured in the correlation between petrological features and remanence polarity for the Mull dykes described by Ade-Hall & Wilson (1969). Examination of Table 1 of the paper describing these correlations shows that no dyke has been subsequently deleted from the normal and reverse groups of dykes whereas two dykes (AO3D155, 172) should be deleted from the shallow reverse group. Recalculation of the values of the magnetite oxidation and granulation numbers and the ilmenite percentage after the removal of the two dykes from the shallow reverse group yields values of 1·034, 1·41 and 1·72 respectively. If these values are plotted on Fig. 2 of the correlation paper it will be seen that no essential change to the correlations results from the deletions.

A small number of baked rocks was subjected to thermal demagnetization as it was thought that alternating field demagnetization might not be capable of separating PTRM, due to baking, from the original TRM of the lava. Demagnetization was carried out using a vector furnace designed by A. E. Mussett (Mussett *et al.* in preparation). Three component measurements of the NRM were made at 50°C intervals until complete thermal demagnetization was achieved.

5. Results of natural remanence measurements

5.1 *General points.* Two general results of these measurements deserve mention before the details are described.

Firstly, the natural remanence of the dyke specimens was characteristically very soft, with 50 Oe field often producing major changes. A stability factor, S, (Wilson *et al.* 1968) has been defined to describe the way in which the NRM of a rock responds to alternating field demagnetization. Thus an NRM where the moment decreases rapidly or the direction swings through nearly 180° will have a stability factor close to zero, while a little affected NRM will have a stability factor close to unity. Fig. 5 shows a comparison of the stability distribution for these dykes (Fig. 5(a)) with the distribution for a collection of sub-aerial lavas (Fig. 5(b)). The dyke rocks are characterized by low S_{200} values (S_{200}: stability factor after demagnetization in 50 Oe steps to 200 Oe). The peak in the distribution is for $0{\cdot}10 < S_{200} < 0{\cdot}20$, the mean value S_{200} is 0·20 and 79·5 per cent of the collection have $S_{200} < 0{\cdot}30$.

In contrast, the lavas have a much broader S_{200} distribution with $\bar{S}_{200}$ at 0·48. The distribution of S_{200} for the baked rocks is also shown in Fig. 5(a); the mean stability, S_{200} at 0·60 is higher than for both dykes and lavas. The explanation of the differences in stability distributions relates to degree of octahedral exsolution in titanomagnetite and appears to be little related to degree of regional hydrothermal alteration. Thus, as is seen in Fig. 6(a), the dykes are characterized by low deuteric oxidation with a low degree of exsolution while the lavas are characterized by a widespread degree of deuteric oxidation. Fig. 6(b) shows this control of stability by oxidation in the form of S versus demagnetizing field plots for different magnetite deuteric oxidation number (M) intervals for a reference collection of Icelandic lavas. The origin of the high average stability of the baked rocks almost certainly lies in the production of masses of fine octahedral exsolution lamellae in initial low deuteric oxidation titanomagnetite as a result of baking. Such fine lamellae are also produced by the laboratory heating of some low deuteric oxidation basalts with concomitant marked increase in stability (Larson *et al.* 1969; Ade-Hall *et al.* 1968 Part III, Plate 1).

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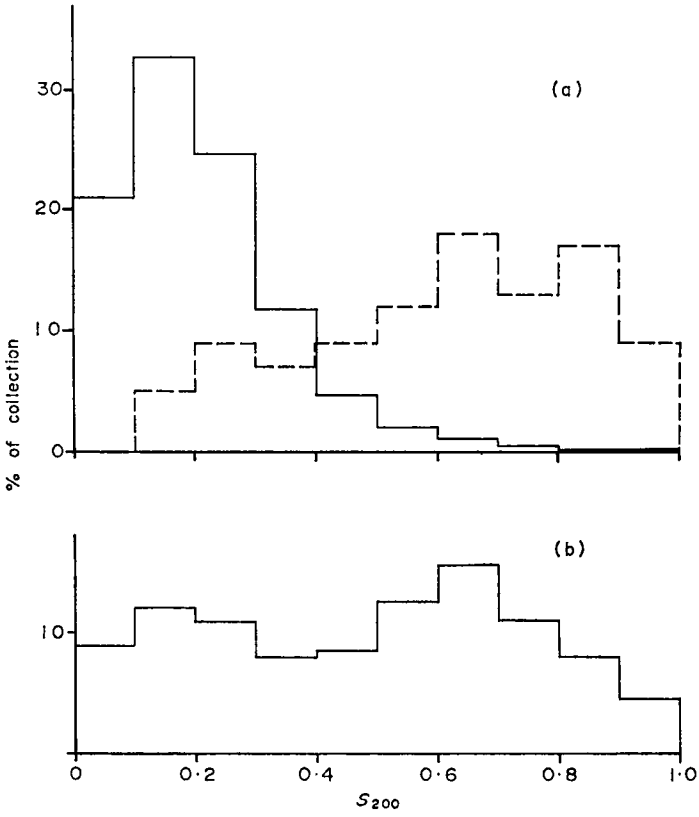


FIG. 5. Distribution of the stability parameter, S_{200}, for different groups of specimens. (a) (———) Mull regional dyke swarm; (– – – – –) baked rocks associated with the Mull dykes; (b) Reference collection of Eastern Iceland basaltic lavas.

The second general result of the alternating field cleaning of the NRMs is that the cleaned directions for each dyke showed a marked improvement in clustering notwithstanding the softness of the natural moments and the large fractions of initial moments removed during the cleaning process. The success of the cleaning method can be judged from the average mean value of the precision parameter α_{95} of 5° for the 154 dyke mean directions listed in Table 1; 96 per cent (148) of the mean directions have $\alpha_{95} < 10°$. While the statistical justification for estimating α_{95} values for only two to four specimens may be questioned the exercise is justified on the grounds that the small values obtained are consistent with the physical situation of accurately defined magnetization directions. The progress of alternating field cleaning for three dykes, with associated baked rocks in two cases, is illustrated in Fig. 7.

5.2 *Dyke cleaned NRM directions* (Fig. 8, Tables 1 and 2).

Both normally and reversely magnetized dykes occur in the collection.

The group of 39 normal dykes is the first large group of normally magnetized rock units recorded from the British Tertiary volcanic province. Dykes and lavas of normal polarity have been reported from Mull by Bruckshaw & Vincenz (1954), Vincenz (1954) and Dutt (1955). However, these workers did not have effective remanence cleaning systems and the possibility exists that their normal polarities may

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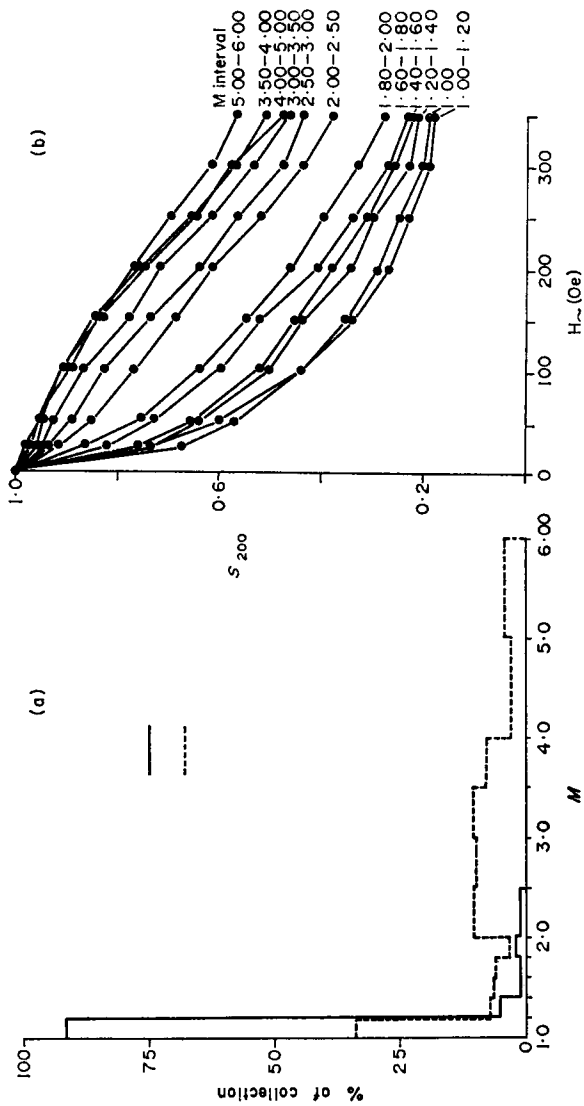


FIG. 6. (a) A comparison of the magnetite deuteric oxidation state distributions for the Mull regional dyke swarm (———), and the Eastern Iceland lavas reference collection (– – – – –); (b) The relationship between stability of remanent magnetization, S, and the titanomagnetite deuteric oxidation state as found in the Eastern Iceland lavas reference collection.

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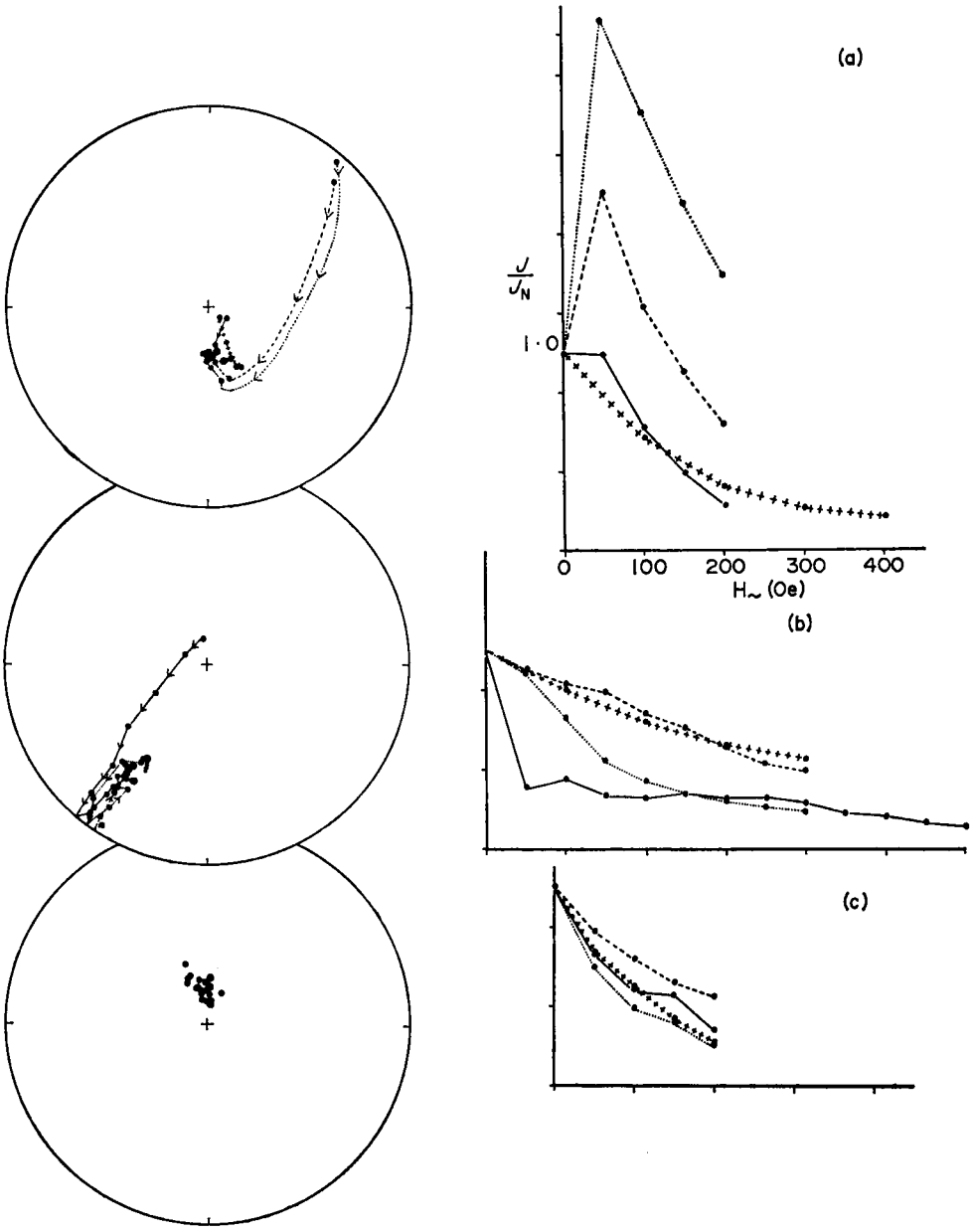


FIG. 7. Examples of the success of the alternating field cleaning of the NRM of material from different dykes. Arrows indicate the direction of movement of the direction on demagnetization. Peak field directions are ringed. (a) Dyke A03D259, representing reversely magnetized dykes (cleaned mean direction: 179, −70, α_{95}: 2). Note the increase in NRM intensity during the initial part of the cleaning process. (+ + + + + +) Baked rock specimen 259A4, remainder dyke specimens; (b) Dyke A03D240, representing shallow reversely magnetized dykes (cleaned mean direction 216, −29, α_{95}: 3) (+ + +) Baked rock specimen 240A, remainder dyke specimens; (c) Dyke A03D175, representing normally magnetized dykes (cleaned mean direction 345, +75, α_{95}: 2). All dyke specimens.

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Table 1

Mean directions, pole positions and divergence angles for individual dykes

Dyke AO3D124 is at the Ardmore extremity of the traverse and AO3D300 at the Quinish extremity. Pole positions are calculated for equivalent normal values of D and I (i.e.) 140, −84 becomes 320, +84. Divergence angle is the angle between a dyke mean direction and the geocentric axial dipole field for the present geographic position of Mull.

Dyke	D	I	α_{95}	Pole	Divergence	Number of specimens collected (number used in obtaining average direction)
AO3D124	140	−84	3	335 °E, 64 °N	14	4(4)
125	176	−64	4	189 , 78	8	4(4)
126	166	−65	3	233 , 77	8	4(4)
126Z*	165	−47	6	201 , 60	26	4(3)
127	163	−64	5	225 , 75	10	4(4)
128	179	−65	3	180 , 80	7	4(4)
129	164	−64	6	223 , 75	10	3(3)
130	178	−69	5	187 , 85	3	4(4)
131	230	−74	5	054 , 65	14	4(4)
132	184	−67	6	152 , 83	5	3(2)
133	171	−59	3	197 , 72	13	4(4)
134	173	−41	5	186 , 56	31	4(4)
135	177	−52	3	180 , 66	20	3(2)
136	180	−39	4	174 , 55	33	4(3)
137	181	−27	7	173 , 48	45	3(2)
138	R or Sh.R		R baked rock			3(0)
139	162	−50	3	207 , 70	23	4(4)
140	191	−35	9	156 , 52	37	3(2)
141	228	−80	5	030 , 65	14	3(3)
142	175	−43	6	183 , 58	29	4(4)
143	212	−63	8	102 , 66	15	3(3)
144	178	−42	9	177 , 58	30	3(3)
145	189	−36	6	160 , 53	36	3(2)
146	R or Sh.R		R baked rock			3(0)
147	180	−51	8	171 , 65	21	4(2)
148	Probably R		R baked rock			3(0)
149	R		R baked rock			3(0)
150	R		R baked rock			3(0)
151	185	−60	6	161 , 74	12	4(2)
152	only one core taken					1(0)
153	168	−48	3	197 , 61	24	2(2)
154	180	−57	3	175 , 71	15	4(2)
155	Polarity uncertain, no baked rock					3(0)
156	157	−75	2	299 , 77	7	4(4)
157	185	−64	6	157 , 78	8	4(3)
158	177	−31	6	178 , 50	41	4(4)
159	184	−73	5	045 , 87	2	4(4)
160	195	−61	6	135 , 72	12	4(4)
161						
162						
163	These numbers not allocated					
164	(Dykes 160 and 166 are adjacent)					
165						
166	Only one core taken, polarity uncertain					1(0)
167	171	−21	4	186 , 44	51	4(4)
168	† sheet, cuts dyke 169 (see Table 4)					3(3)

*Z This suffix 'Z' indicates that the dyke was omitted from the first numbering scheme.

† sheet-dyke intersections are used to find the age-relationships of normal, reverse and shallow reverse magnetic directions (Table 4).

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Table 1 (*continued*)

Dyke	*D*	*I*	α_{95}	Pole	Divergence	Number of specimens collected (number used in obtaining average direction)
169	186	−49	3	172 , 63	23	4(4)
170	172	−44	4	187 , 59	28	3(3)
171	210	−58	8	114 , 63	18	3(2)
172	Probably Sh.R, no baked rock					4(0)
173	176	−36	3	181 , 53	36	3(3)
174	157	−56	7	220 , 65	18	2(2)
175	343	+75	2	294 , 80	6	4(4)
176	R or Sh.R		R	baked rock		3(0)
177	181	−25	5	172 , 46	47	3(3)
178	196	−38	6	148 , 53	35	4(4)
179	146	−40	8	225 , 48	36	3(3)
180	179	−61	7	175 , 76	11	4(4)
181	126	−59	9	262 , 52	25	3(3)
182	181	−58	9	170 , 73	14	3(2)
183	190	−65	7	136 , 79	8	3(2)
184	167	−58	2	204 , 70	15	4(4)
185	160	−24	11	201 , 43	49	3(2)
186	343	+77	6	320 , 78	7	3(2)
187	158	−64	13	254 , 72	11	3(3)
188	180	−84	7	354 , 69	12	3(3)
189	192	−63	6	138 , 76	10	3(3)
190	sheet, cuts dyke 189 (see Table 4)					4(4)
191	170	−61	9	203 , 74	11	3(2)
192	323	+64	5	255 , 65	16	3(3)
193	179	−58	3	176 , 72	14	3(3)
194	158	−61	4	228 , 70	14	3(3)
195	180	−64	5	174 , 79	8	4(4)
196	171	−64	7	206 , 78	8	3(2)
197	267	−85	4	012 , 56	19	4(4)
198	224	−83	3	018 , 65	14	4(4)
199	205	−71	6	079 , 76	8	4(4)
200	192	−70	3	094 , 83	4	3(3)
201	161	−71	4	262 , 79	6	3(3)
202	sheet, cuts dyke 203 (see Table 4)					4(4)
203	178	−62	8	181 , 63	10	4(4)
204	R or Sh.R		R	baked rock		3(0)
205	357	+63	2	182 , 77	9	2(2)
206	338	+69	5	254 , 77	8	4(4)
207	355	+67	2	201 , 83	5	4(4)
208	349	+67	2	221 , 80	6	3(3)
209	169	−53	3	196 , 66	19	3(3)
210	192	−45	6	153 , 59	27	4(4)
211	157	−40	2	210 , 52	34	4(3)
212	157	−54	1	218 , 63	20	4(4)
213	172	−68	4	222 , 83	5	4(4)
214	sheet					3(3)
215	154	−74	5	288 , 76	8	3(2)
216	178	−63	3	179 , 77	9	4(4)
217	282	−50	5	057 , 19	47	4(4)
218	178	−30	5	177 , 50	42	3(3)
219	139	−64	4	259 , 63	17	4(4)
220	182	−64	2	166 , 79	8	4(4)
221	188	−68	5	128 , 83	5	4(4)
222	152	−60	8	253 , 66	16	3(3)
223	169	−65	3	212 , 78	8	4(4)
224	102	−80	6	320 , 56	19	4(2)
225	192	−41	2	155 , 56	31	4(4)
226	007	+66	2	141 , 81	6	4(4)

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Dyke	D	I	α_{95}	Pole	Divergence	Number of specimens collected (number used in obtaining average direction)
227	171	−41	3	189 , 57	31	4(4)
228	180	−58	4	174 , 74	14	4(3)
229	198	−61	9	128 , 71	13	2(2)
230	178	−59	3	180 , 73	13	4(4)
231	208	−42	4	130 , 52	33	4(4)
232	198	−71	5	079 , 80	6	4(4)
233	184	−66	7	155 , 81	6	4(4)
234	203	−73	3	066 , 78	7	4(4)
235	Sh.R					4(0)
236	201	−64	5	114 , 73	11	3(3)
237	203	−74	8	058 , 78	7	4(4)
238	024	+74	3	054 , 77	7	3(3)
239	198	−66	7	111 , 76	9	3(2)
240	216	−29	3	125 , 41	47	3(3)
241	217	−53	3	110 , 56	25	3(3)
242	177	−55	6	181 , 69	17	3(3)
243	187	−53	3	178 , 67	19	4(4)
244	Only one core taken (R)					1(0)
245	168	−67	5	227 , 80	6	3(3)
246	170	−55	7	196 , 68	17	4(4)
247	172	−63	3	200 , 77	9	3(3)
248	013	+62	1	137 , 75	11	3(3)
249	176	−23	21	180 , 45	49	3(2)
250	324	+75	2	296 , 71	11	3(3)
251	005	+70	3	128 , 86	2	3(3)
252	309	+69	8	278 , 61	17	3(3)
253	sheet, cut by dykes 252, 254 (Table 4)					3(3)
254	349	+77	3	328 , 80	6	3(3)
255	297	+76	5	304 , 59	17	3(3)
256	321	+79	6	315 , 69	12	3(3)
257	014	+68	7	110 , 80	6	3(3)
258	185	−50	7	168 , 64	22 } in same	4(4)
259	179	−70	2	184 , 87	2 } fissure	3(3)
260	Polarity uncertain, N baked rock					3(0)
261	201	−20	13	146 , 41	53	3(3)
262	333	+82	4	333 , 69	12	3(3)
263	324	+78	2	309 , 71	11	4(4)
264	333	+68	3	255 , 73	10	3(3)
265	174	−47	2	184 , 61	25	3(3)
266	336	+74	2	293 , 77	7	4(4)
267	316	+76	2	299 , 68	13	4(4)
268	182	−59	3	179 , 73	13	3(3)
269	182	−45	6	163 , 60	27	4(4)
270	304	+75	5	298 , 62	16	3(3)
271	288	+75	5	320 , 58	20	3(3)
272	307	+70	3	283 , 58	17	4(4)
273	301	+55	3	262 , 46	29	3(3)
274	283	+81	2	321 , 57	18	3(3)
275	154	−65	5	255 , 68	12	3(3)
276	305	+81	3	322 , 63	15	3(3)
277	173	−48	8	187 , 62	24	3(3)
278	132	−87	7	344 , 61	16	4(4)
279	153	+88	7	357 , 53	20	3(3)
280	327	+80	2	321 , 70	11	3(3)
281	014	+75	4	042 , 81	5	3(3)
282	174	−71	1	242 , 86	2	3(3)
283	179	−56	4	176 , 70	16	3(3)
284	194	−00	10	157 , 32	72	3(3)
285	193	−02	5	158 , 33	70	3(3)
286	326	+78	3	313 , 71	11	3(3)

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Table 1 (*continued*)

Dyke	D	I	α_{95}	Pole	Divergence	Number of specimens collected (number used in obtaining average direction)
287	216	−64	3	094 , 66	15	4(4)
288	190	−75	8	110 , 77	4	4(4)
289	158	−67	7	245 , 75	9	4(4)
290	345	+89	3	353 , 58	17 } in same	4(2)
291	207	−75	4	051 , 75	8 } fissure (Table 4)	4(4)
292	338	+79	3	323 , 75	9	4(4)
293	216	+86	8	347 , 51	22	4(3)
294	195	−69	4	099 , 81	6	4(4)
295	295	+59	4	277 , 41	28	4(4)
296	317	+73	7	288 , 68	13	4(4)
297	027	+76	14	042 , 75	8	4(4)
298	312	+77	5	304 , 66	13	4(4)
299	206	−79	1	027 , 73	10	4(4)
300	159	−69	14	254 , 77	8	4(4)

Table 2

Mean directions of magnetization for groups of dykes and angles of divergences of mean directions from the geocentric axial dipole field for the present geographic latitude of Mull

Mean directions

Group	Number	$\bar{D}$	$\bar{I}$	k	α_{95}	Pole position	Errors on mean pole (δp, δm)
All dykes	154	356	+63	17	2·7	187, 78°N	3, 4
All normal dykes	39	334	+76	54	3·0	301, 75	5, 6
All reverse and shallow reverse dykes	115	180	−59	18	3·1	174, 73	3, 4

Divergence angles:

Divergence angle between mean direction for all dykes and the geocentric axial dipole field for the present geographic latitude of Mull: 8°.

Divergence angle between the mean direction for the group of normal dykes and the geocentric axial dipole field for the present geographic latitude of Mull: 8°

Divergence angle between the mean direction for the group of reverse and shallow reverse dykes and the geocentric axial dipole field for the present geographic latitude of Mull: 13°

Divergence angle between normal and reverse/shallow reverse group axes: 19°

have been the result of secondary magnetization acquired in the present field. On the other hand, Raja (1964) and Smith (1966), who used remanence cleaning techniques, have each found in Ayrshire, Scotland, a single normally magnetized Tertiary dyke.

The remaining 115 dyke mean directions form an extended group all having southerly declinations with inclinations ranging from −87° to the horizontal. While this group contains some steep reverse directions almost exactly opposing the geocentric axial dipole field for the present latitude of Mull, there is no clear distinction between these steep reverse directions and the abundant shallower directions. However, for comparisons within this group we sub-divide it into reverse dykes ($-90 < I \leqslant -55$)

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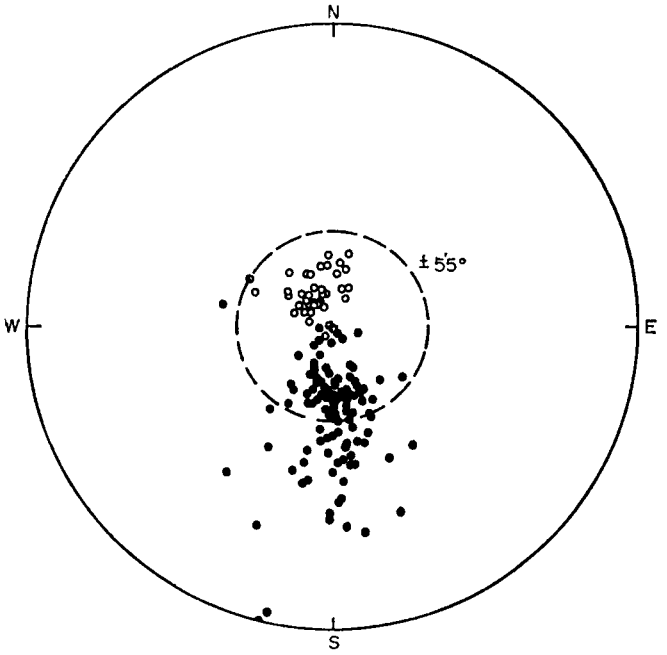


FIG. 8. Distribution of dyke mean directions of magnetization. (Equal angle stereographic plot), open circles, lower hemisphere; full circles, upper hemisphere.

and shallow reverse dykes ($-55 < I$). The division at an inclination of $-55°$ is selected as the shallowest normal dyke has an inclination of $+55°$.

The clear asymmetry of the normal and 'reverse' groups of mean directions is surprising in the light of the conventional picture of the geomagnetic field as having two antiparallel stable states. This assymetry must form a major point in discussing the possibilities of partial remagnetization.

5.3 *Baked rock cleaned NRM directions* (Fig. 9 and Table 3)

In this section we will be concerned with the agreement of dyke mean and baked rock cleaned direction of magnetisation as this factor must be of importance in assessing the originality of dyke cleaned NRM directions.

5.3.1 *Baked rocks defined.* A total of 106 oriented country rock specimens were collected during the drilling programme. Since the position of country rock specimens relative to dyke contacts together with dyke thicknesses vary widely, a criterion has to be chosen to limit the number of country rock specimens which may be expected to show NRM directional agreement with their adjacent dyke. The initial criterion selected is that 'baked rock' specimens must have been located at a distance from the dyke-country rock contact of less than a tenth of the width of the dyke. Another factor which could obscure directional agreement between dyke and baked rock NRMs is a poorly-defined dyke cleaned NRM direction. Thus we exclude the few baked rocks where the baking dyke has an α_{95} of greater than 10°. The reliability of these criteria is discussed later. However, experience suggests that we may expect many specimens within the limit to have been completely remagnetized. In all 70 country specimens, being baked by 57 dykes, form this circumscribed group. These are indicated by an asterisk in Table 3 and are called 'baked rocks' in contrast to 'country rocks'.

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5.3.2 *Characteristics of baked rock NRMs.* Several features characterize this group of baked rocks.

Firstly, polarity agreement exists for all but one baked rock. That is, with one clear exception, and several cases of bad agreement, normal dykes, shallow reversed dykes and reversed dykes have clearly normal, shallow reversed and reversed baked rocks respectively. The single case of polarity disagreement and five cases of bad agreement are discussed individually later.

Secondly, examination of the level of agreement between dykes and baked rocks in each of the normal, shallow reverse and reverse groups shows that agreement is generally not as good as would be expected from a simple model of dyke and baked rock acquiring identical directions of magnetization by magnetization in the same ambient field. This is clearly seen in Fig. 9. Both the linear relationship between dyke and baked rock inclinations (with a correlation coefficient of +0·95) and the degree of scatter are evident. Alternatively the level of disagreement is seen by comparing the α_{95} angles for dykes with the divergence angles (Blackett, Clegg & Stubbs 1960) between dyke mean and baked rock directions of magnetization (Table 3). The divergence angle is equal to or less than α_{95} for only 17 (30 per cent) of the 57 dykes.

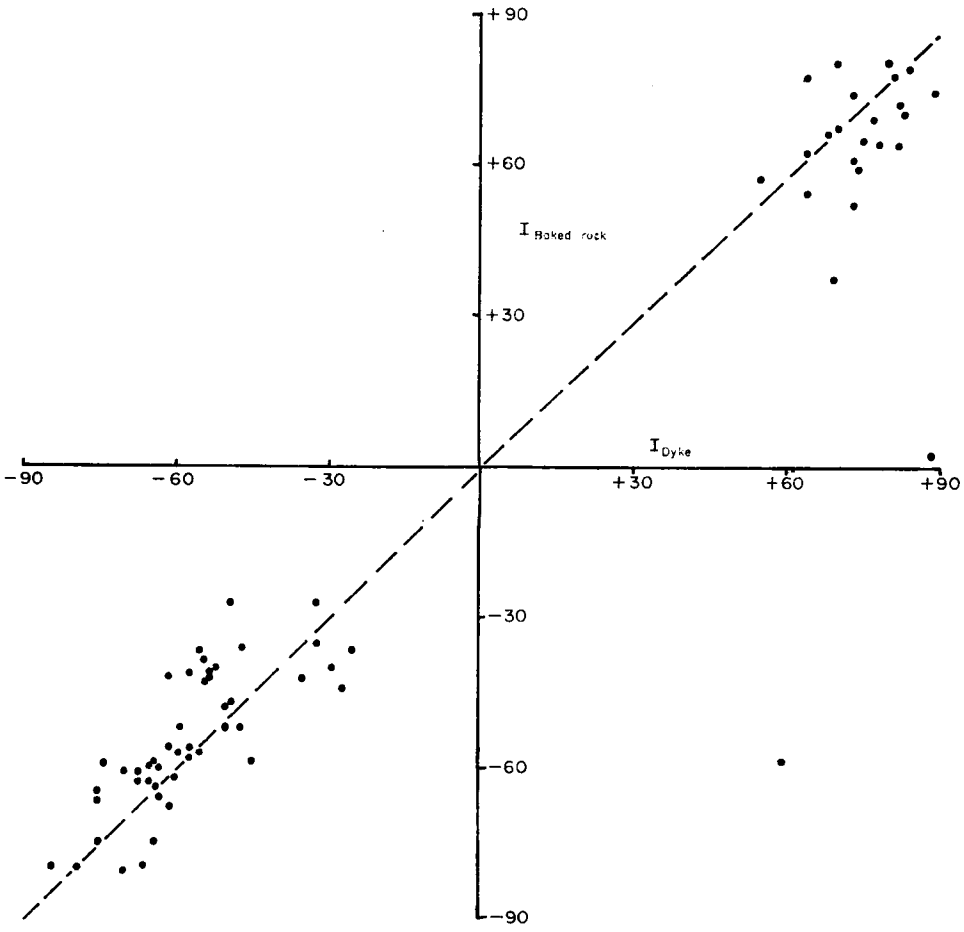


FIG. 9. Baked rock cleaned NRM inclination plotted against baking dyke mean cleaned NRM inclination. The diagonal line defines the locus of points for which baked rock and dyke inclination are identical.

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Table 3

Country rock directions of cleaned magnetization

Country rock directions are given with associated dyke directions. Divergence angles are angles between dyke and country rocks directions of cleaned magnetization.

(Baked rocks, as distinct from country rocks in general, are defined as being separated from the contact by less than one tenth of the dyke width. Those baked rocks within the distance associated with dykes for which $\alpha_{95} < 10°$ are marked with an asterisk).

Number	Dyke		Width (cm)	Country Rocks				Distance from contact in cm (fract distances in brackets)
	D	*I*		Letter	*D*	*I*	Div.	
AO3D129	164	−62	22	A*	176	−60	7	1 (0·05)
132	184	−67	uncertain	A	152	−62	15	4
134	171	−43	300	A*	106	−54	45	4 (0·01)
135	177	−52	34	A*	182	−47	6	2 (0·06)
137	181	−27	180	D*	172	−49	23	3 (0·02)
				E*	179	−32	5	3 (0·02)
138	R or Sh.R		75	A	163	−67	—	3 (0·04)
140	191	−35	70	B*	183	−32	7	3 (0·04)
141	228	−80	75	A*	195	−79	6	0 (0·00)
143	212	−63	22	A	182	−67	13	4 (0·18)
144	178	−42	100	A*	183	−35	8	2 (0·02)
145	189	−36	100	A*	214	−55	26	3 (0·03)
146	R or Sh.R		160	A	186	−57	—	3 (0·02)
148	Probably R		160	A	176	−46	—	3 (0·02)
				B	175	−54	—	30 (0·16)
149	R		100	A	143	−60	—	5 (0·05)
150	R		40	A	163	−59	—	2 (0·05)
				B	176	−67	—	3 (0·08)
170	172	−44	45	A*	159	−27	20	4 (0·09)
171	210	−58	63	A*	205	−57	3	3 (0·05)
173	176	−36	190	A*	178	−25	11	2 (0·01)
174	157	−56	200	A*	156	−57	1	0 (0·00)
				B*	159	−61	5	0 (0·00)
				C	158	−53	3	22 (0·11)
				D	153	−54	3	38 (0·19)
176	R or Sh.R		55	A	184	−58	—	2 (0·04)
179	146	−40	70	A*	136	−52	14	3 (0·04)
181	126	−59	55	A*	139	−74	16	0 (0·00)
186	343	+77	440	A*	328	+64	14	0 (0·00)
187	158	−64	100	A	186	−59	14	5 (0·05)
188	180	−84	170	A*	160	−80	5	3 (0·02)
189	192	−63	95	A*	190	−66	3	3 (0·03)
191	170	−61	430	A*	241	−42	45	3 (0·01)
				B*	171	−68	7	6 (0·01)
192	323	+64	210	A*	305	+62	9	0 (0·00)
				B*	307	+54	13	0 (0·00)
193	179	−58	145	A*	186	−57	4	3 (0·02)
				B*	191	−64	8	4 (0·03)
194	158	−61	40	A	169	−68	8	5 (0·13)
196	171	−64	46	A*	172	−75	11	3 (0·07)
200	192	−70	24	A*	193	−61	9	0 (0·00)
209	169	−53	uncertain	A*	160	−41	14	0 (0·00)
218	178	−30	40	A	185	−20	12	5 (0·13)
236	201	−64	40	A*	209	−64	4	3 (0·08)
238	024	+74	15	A*	348	+59	20	0 (0·00)
239	198	−66	56	A*	247	−80	19	4 (0·07)
240	216	−29	910	A*	211	−40	12	2 (0·00)
241	217	−53	100	A*	215	−42	11	3 (0·03)
242	177	−55	80	A*	161	−57	10	3 (0·04)
245	168	−67	28	A*	153	−61	9	2 (0·07)
247	172	−63	30	A*	168	−60	4	2 (0·07)

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Table 3 (*continued*)

Number	Dyke D	Dyke I	Width (cm)	Country Rocks Letter	D	I	Div.	Distance from contact in cm (fract distances in brackets)
248	013	+63	35	A	341	+58	16	6 (0·17)
250	324	+75	115	A*	342	+65	12	4 (0·04)
251	005	+70	70	A	356	+66	6	38 (0·54)
252	309	+69	450	A*	112	+37	74	3 (0·01)
254	349	+77	60	A*	325	+69	10	0 (0·00)
255	297	+76	150	A*	307	+78	3	4 (0·03)
256	321	+79	170	A*	353	+79	6	4 (0·02)
257	014	+68	135	A*	020	+66	3	3 (0·02)
258	183	−50	150	A*	187	−59	9	2 (0·01)
				B*	176	−52	6	4 (0·03)
				C*	119	−48	36	0 (0·00)
259	179	−70	160	A*	154	−63	12	6 (0·04)
								Baked 258
				B*	121	−81	17	3 (0·02)
260	Polarity uncertain		270	A	112	+74	—	3 (0·01)
261	201	−20	220	A	184	−38	25	0 (0·00)
262	333	+82	64	A*	335	+72	10	0 (0·00)
263	324	+78	260	A*	249	+64	25	2 (0·01)
264	333	+68	700	A*	000	+74	11	3 (0·00)
265	174	−47	90	A*	176	−36	11	0 (0·00)
268	182	−59	60	A*	201	−52	13	0 (0·00)
270	304	+75	50	A	234	+66	23	33 (0·66)
271	288	+75	70	A	275	+71	5	8 (0·11)
273	201	+55	67	A*	297	+57	3	0 (0·00)
				B	170	−61	154	130 (1·94)
274	283	+81	70	A*	209	+80	12	6 (0·09)
275	154	−65	130	A*	154	−60	5	7 (0·05)
276	305	+81	15	A	298	+66	15	2 (0·13)
277	173	−48	570	A*	167	−47	4	9 (0·02)
279	153	+88	340	A*	148	+04	84	0 (0·00)
280	327	+80	240	A*	307	+81	4	3 (0·01)
281	014	+75	170	A*	038	+67	11	4 (0·02)
282	174	−74	16	A	170	−72	2	5 (0·31)
282	179	−56	200	A*	187	−41	16	0 (0·00)
284	194	00	67	A	195	−06	6	3 (0·04)
285	193	−02	10	A*	192	−19	17	0 (0·00)
286	326	+78	340	A*	312	+70	9	0 (0·00)
288	190	−75	33	A*	190	−75	0	3 (0·09)
				B	207	−70	8	6 (0·18)
				C	209	−76	5	350 (10·6)
				D	229	−67	15	410 (12·4)
290	345	+89	300	A*	039	+74	15	0 (0·00)
				B*	034	+74	15	0 (0·00)
				C	037	+80	9	169 (0·56)
				D	032	+76	13	305 (1·02)
291	207	−75	81	A*	193	−67	10	0 (0·00)
								Baked 290
				B*	191	−65	12	0 (0·00)
295	295	+59	160	A*	193	−59	142	2 (0·01)
296	317	+73	190	A*	339	+52	23	7 (0·04)
				B*	346	+61	16	15 (0·08)
				C	152	+82	35	32 (0·17)
				D	007	+86	15	59 (0·31)
				E	092	+16	91	107 (0·56)
				F	180	−46	147	178 (0·94)
				G	231	+43	48	321 (1·68)
				H	273	−78	153	650 (3·42)

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The problem then is to explain, and if possible assign a significance, to this level of dyke-baked rock directional disagreement.

There are several possible reasons for the level of dyke-baked rock directional agreement.

(i) Undetected core orientation and core-marking errors. These will be hard to recognize since with generally only a single-baked rock per dyke a test for internal consistency of baked rock directions cannot be applied.

(ii) Failure of baked rock selection criteria. Selection of criteria to distinguish baked rocks from the remaining country rock specimens was in the ultimate arbitrary and took no account of the expected spatial variation of significant basalt properties such as Curie point. However, several additions to the selection criteria based on examination of the magnetic minerals of the baked rocks and other features failed to produce groupings of baked rocks in significantly better directional agreement with their dykes. One feature which must be considered as a possible source of directional disagreement is the narrow estimated width of many completely remagnetized zones. Thus, a conservative estimate of the width of completely remagnetized zones, as a fraction of dyke width, suggests that one third of completely remagnetized zones will have widths just equal to or less than the diameter of a drill core specimen. Thus many assumed baked specimens may be expected to contain a fraction of only partly remagnetized material.

In the context of failure of selection criteria it is illuminating to see the dyke-baked rock divergence histograms for normal, shallow reverse and reverse groupings (Fig. 10). The mean divergence is least for the reverse pairs and greatest for the normal pairs. This sequence is consistent with dyke intrusion into reversely magnetized lavas with different degrees of remagnetization of contact rocks. That is, incomplete remagnetization of a reversely magnetized lava in a reverse field would result in only small divergences while divergence would increase successively for incomplete remagnetization of reversely magnetized lava when the ambient field had respectively a shallow reverse or normal direction. Some of the evidence may favour such an interpretation: while the few country rock specimens collected during the dyke sampling do not show a constant reverse direction of magnetization (Table 3) the Mull lava pile, where sampled 25 km away at its thickest remaining development, is entirely reversely or shallow reversely magnetized (Ade-Hall *et al.* 1969).

(iii) Failure of a.f. demagnetization to separate PTRMs. Thermal demagnetization of the NRMs of four baked rocks was carried out to test for this possibility. No clear evidence of the presence of low temperature PTRMs was found, this result being consistent with the results of a.f. demagnetization.

(iv) Post-baking magnetization processes. Here we are concerned with the possibilities of post-baking magnetization changes in the baked rocks; a similar consideration of dyke post intrusion magnetization changes is given in Section 6.

There are two important facts to consider here, the first deals with evidence for post-baking changes to the opaque mineralogy of the baked rocks and the second with the dyke-baked rocks pairs having divergences in excess of 30°.

Titanomagnetite granulation is seen in some of the low deuteric oxidation baked rocks, as it is in many of the predominantly low deuteric oxidation dykes. However, baked specimens containing granulated titanomagnetite do not have systematically higher divergences from their respective dykes than do specimens with ungranulated low deuteric oxidation magnetites. This suggests that the titanomagnetite alteration process leading to granule formation does not cause directional remagnetization, a conclusion in line with the result of the fuller discussion of this possibility in Section 6.

For the six dyke-baked rock pairs where the divergence exceeds 30° the possibility must be considered that either the dyke or the baked rock of a pair has been remagnetized at some time since the cooling of the dyke. Of the six dyke-baked rock pairs with

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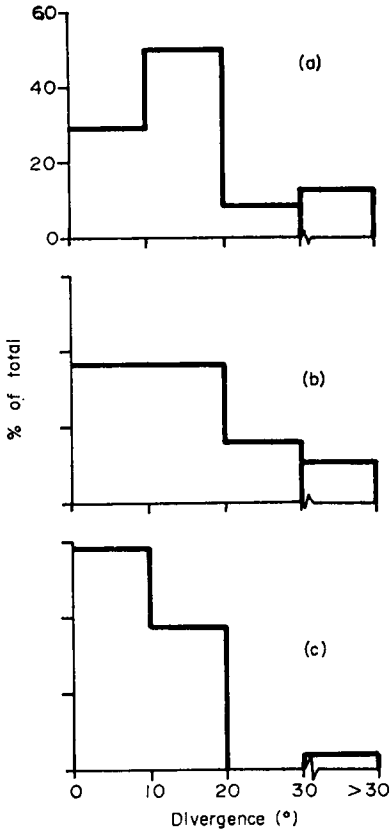


FIG. 10. Baked rock divergence angle distributions. (a) Baked rocks for normal dykes; (b) Baked rocks for shallow reverse dykes; (c) Baked rocks for reverse dykes.

divergences of greater than 30°, two (AO3D134A, 258C) for shallow reverse dykes may owe their large divergences to field errors. Thus 134A was one of three rather similarly oriented baked rocks of which the other two are no longer available for study. Confusion may have taken place in core numbering immediately after drilling. Again, the field record for 258C indicates that it was broken off during drilling and there was some doubt about the refit position. Also in favour of a field error for 258C is the fact that 258A and 258B, situated slightly further from the dyke than 258C (and thus not conforming with the numbering system described in Section 3) have relatively small divergence angles of 9° and 6° respectively. A field error may also explain the large divergence of reverse baked rock 191A as 191B, situated further from the dyke that 191A, has the relatively small divergence angle of 7°. However, there is no record in the field notes which might explain the large divergence angles of the normal baked rocks 252A, 279A and 295A. Here the explanation appears to lie between failure to completely remagnetize on baking or later remagnetization. Summarizing, it seems that the evidence for post-baking remagnetization is distinctly limited but the possibility cannot be ruled out in every case.

5.3.3 *Baked rocks—summary and conclusions.* There are probably sufficient physically reasonable explanations for the level of dyke-baked rock directional disagreement without having to take recourse to the concept of later partial remagnetization, although this possibility cannot be excluded completely.

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Basalt forms a relatively bad country rock in this context, being generally strongly magnetized with Curie points probably ranging rapidly spatially between about 200 and 600°C. Thus spatially uniform reheating will produce different degrees of remagnetization from point to point. The effect of variability in remagnetization will be accentuated if the field during remagnetization is sometimes parallel to and sometimes opposed to the original magnetization direction of the country rock. Attempts to define a generally applicable criterion for guaranteed complete remagnetization suggest that specimens should be taken at not more than one-twentieth of the dyke width from the contact. Since many of the Mull dykes are narrow this will lead to narrow completely remagnetized zones across which high temperature gradients existed during baking.

6. A discussion of the originality of the dyke mean cleaned NRM directions

In this section the evidence for and against the original cooling TRM nature of the cleaned mean NRM directions is listed and discussed and a conclusion presented.

6.1 *Evidence for the original cooling TRM nature of the cleaned NRMs*

A property which the original TRM of a small rapidly cooled volcanic unit must have is minimal directional variation throughout the unit. This must be so as the unit cools in a time interval which is short compared with the time required for significant variation in the ambient field. It is conceivable that the ambient field might be locally distorted by the presence of adjacent relatively strongly magnetized basalt lava country rock, but measurement of recent lavas (Doell & Cox 1963) show that such distortions are only likely to be comparable with the errors of a few degrees accumulated during sampling and measurement. We have shown in Section 5.1 that a.f. cleaned NRM directions almost always have this required within-dyke consistency. Thus, the cleaned NRM directional data is often consistent with an original cooling TRM origin but the possibility of later *uniform* remagnetization must still be considered.

Several lines of evidence indicate that it is unlikely that uniform regional remagnetization has taken place: both normal and reverse directions of cleaned magnetization occur and dykes having these very different directions of magnetization are often closely interleaved. This is best seen in the kilometre interval across the strike of the swarm in which dykes 265 to 300 occur. Here 18 small groups of dykes (nine normal and nine reverse or shallow reverse) alternate. Again, lava baked rock directions, with rare exceptions, are consistent with dyke cleaned directions although the range in divergence angle for each group may deserve further attention. Thus normal baked rocks are associated with normal dykes and reverse and shallow reverse with reverse and shallow reverse dykes respectively. Both this interleaving of N and R dykes and the dyke-baked rock directional agreement are inconsistent with a *regional* alteration process resulting in new magnetic directions. In the light of this conclusion the few baked rocks which show really bad agreement with dyke cleaned NRM directions may not have been completely thermally remagnetized during baking (Section 5.3.3). Comparison of the palaeomagnetic results from the Mull regional dyke swarm with results from other sets of volcanic rock of similar age allows a further test of the original TRM nature of NRMs. For example, the Isle of Skye (Scotland) regional dyke swarm sampled the Vaternish peninsula (57° 34′ N, 353° 20′ E), which is likely to be of similar age to the Mull regional swarm, contains normal, shallow reverse and reverse dykes in proportions similar to those for the Mull swarm. The important point is that the Vaternish dykes have experienced considerably less regional hydrothermal alteration than the Mull dykes. The Curie point distribution and secondary mineral content for the Skye material suggests only zone B burial conditions with a maximum reheating temperature of no more than 100°C. The similar distributions

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of palaeomagnetic directions for the two dyke collections implies that regional reheating and hydrothermal alteration do not obscure original TRM directions to such an extent that they may not be recovered by a.f. cleaning.

6.2 *Evidence against the original cooling TRM nature of the cleaned NRMs*

The two disconcerting features of the dyke material are the widespread moderate replacement of titanomagnetite by ferri-rutile granules (granulation) and the general Curie point rise of between 100 and 300°C above the estimated original Curie points. Both of these features have resulted from regional hydrothermal alteration and must be considered as possible indicators of remagnetization.

The influence of granulation on original TRM direction has been investigated by Ade-Hall *et al.* (1968). In this investigation two groups of specimens from within a single lava flow were used, the first consisting of low deuteric oxidation, high granulation specimens and the second of high deuteric oxidation specimens from which granulation was absent. The precisely defined mean cleaned NRM direction for the groups were not significantly different. It was concluded from these results that the formation of ferri-rutile granules from titanomagnetite had not resulted in remagnetization as the apparently unaltered high deuteric oxidation specimens, with 500 to 600°C Curie points, must have retained the original TRM directions.

The influence of general Curie point rise can be assessed using the dyke swarm comparison information given in the last paragraph of Section 6.1. If two dyke swarms of similar age and very different degrees of burial alteration yield similar palaeomagnetic results, it can be concluded that Curie point rise does not obscure original cooling TRM directions.

Other related features of the dyke swarm also deserve mention in this section. These concern the correlation of opaque petrological properties with polarity in the dykes as reported by Ade-Hall & Wilson (1969). It was found that average titanomagnetite deuteric oxidation state and granulation and separate ilmenite content increased from normal dykes through shallow reversed dykes to reversed dykes. Of these three correlations those involving titanomagnetite deuteric oxidation and separate ilmenite content had been reported for other collections and convincing reasons were given for believing that remagnetization had *not* taken place. The correlation involving titanomagnetite granulation and reverse polarity, seen by comparing Fig. 3(b) and (c), has yet to be explained. However, the interpretation of evidence described in this section and in Section 6.1 is all against loss of original direction during remagnetization and it may be that this relationship between reverse polarity and degree of hydrothermal alteration is coincidental.

Another clear example of asymmetry between normal and reverse units, in this case forming two successive groups in a basaltic lava pile, has been described by Palmer (1970). The lava sequence in question is the Keweenawan of the Lake Superior area. Palmer has shown by critical examination of conglomerate test results that for these lavas, which have been regionally hydrothermally altered to a considerably greater extent than the Mull dykes, the asymmetry in NRM directions is consistent with partial remagnetization of the whole sequence. However, Palmer was unable to show the present of two component magnetizations by high field a.f. analysis of remanences.

6.3 *Conclusion on the originality of the cleaned NRM directions*

The weight of the evidence is in favour of the cleaned NRM directions being at least very close to the original cooling TRM directions. Several classical tests are satisfied in this respect while there is also other evidence to show that a high degree of regional hydrothermal alteration has not produced any significant effects on cleaned remanence directions. An interesting feature, which deserves further investigation is the statistical relationship between the distribution of polarities and the variation in

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degree of hydrothermal alteration within the section. However the possibility cannot be ruled out that some directional change has taken place during hydrothermal alteration, resulting in the assymetrical distribution of direction and the different degrees of scatter of the normal and reverse/shallow-reverse groups. If such directional changes can be positively identified by some means than considerable doubt will be case on the assumed ability of alternating field and thermal remance analysis to separate magnetizations acquired at different times in the history of a rock.

7. The Palaeogene geomagnetic field recorded by the Mull regional dyke swarm

It is clear from the directional data listed in Tables 1 and 2 and illustrated in Fig. 8 that no simple picture consisting of conventional symmetrically opposed N and R geomagnetic field directions is recorded by the dykes. Instead a fairly tight cluster of 39 N directions is matched by a meridionally elongated distribution of 115 directions, extending continuously from conventional R directions to south and horizontal directions. No simply obtained estimate of continental or polar movement could reasonably be obtained from this data.

Ideally, before considering what these asymmetrical distributions mean in terms of geomagnetic field behaviour it would be necessary to find the time sequence of dyke intrusion. This, however, is at the present, impossible to determine for the collection as a whole. In contrast with lavas, there is no generally applicable way to determine the sequence of intrusion of dykes from field observations. The occasional intersections of dykes, the multiple intrusions of two or more dykes in the same fissure, or the intersections of dykes and contemporaneous small basaltic sheets also intruded in the lava country rock can be used for relative age determination. The general K/A dating of the dyke swarm has yet to be carried out and even if data were available it is doubtful, with the present state of the art, whether sufficiently small errors could be obtained in practice to resolve the ages of the different dykes with certainty. Thus we are limited to using the small number of contact relationships found in the field and listed in Table 4.

Table 4

Dyke and approximately contemporaneous basaltic sheets observed in the field to be in contact (N: normal, Sh. R: shallow reverse, R: reverse)

Older unit		Younger unit(s)
AO3D	128(R)	126(R), 127(R)
	169(R)	168(R)
	189(R)	190(R)
	203(R)	202(Sh.R)
	253(Sh.R)	252(N), 254(N)
	258(R)	259(R)
	290(N)	291(R)

The results listed in this table do not suggest a simple time sequence for the geomagnetic field changes. Thus, using the three pairs involving different palaeomagnetic direction groups we have

for 203/202	$R > Sh.R$	(i.e. a reverse field preceding a shallow reverse field)
for 253/252, 254	$Sh.R > N$	
and 290/291	$N > R$	

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The reasonable interpretation of this very limited amount of evidence is that the dykes and sheets record parts of at least three polarity epochs, two reversed and an intervening normal.

Another problem peculiar to a dyke swarm study is that the branching of intrusions may lead to different dykes being taken erroneously as separate intrusions. This situation can sometimes be recognized with reasonable certainty, as in the case of the two nearly horizontally magnetized dykes 284 and 285, which are only separated where sampled by a few meters of lava and the group of adjacent normal dykes from 205 to 208. However, where branching has taken place well below the present ground surface, the branches at the surface may be hundreds of meters apart and quite unrecognizable as being parts of the same intrusion. Thus duplication of data may be expected in the palaeomagnetic results.

Bearing in mind the difficulties in determining the time order and true number of independent determinations of the ambient field direction recorded by the dykes, what can usefully be obtained from the palaeomagnetic data? Several aspects of the data can be considered profitably and general theories concerning the volcanic activity and the Palaeogene magnetic field can be presented for testing in the future.

The most interesting feature of the palaeomagnetic data is the meridional spread of southerly oriented directions. A simple interpretation of the spread is that rapid dyke intrusion has resulted in the recording of part of a polarity transition. In the absence of a knowledge of the time order of intrusion of the dykes some other means must be found of testing this hypothesis. A method devised by Dagley *et al.* (1967) can be used. Cleaned NRM intensities for N, R and A (Anomalous or transition) directions were compared for Icelandic lavas and Dagley *et al.* found that twice as many A specimens had $J_{150} < 5 \times 10^{-4}$ emu g^{-1} as did N or R specimens, so distinguishing the transition group. The abundant low intensities found for A lavas suggests that the geomagnetic field is considerably reduced during transitions, and this has been confirmed for the Icelandic lavas by palaeomagnetic field intensity measurements made by Lawley (1970). Application of this test to the Mull dykes does not yield such a clear picture although this may be because only part of a transition was recorded: 53 per cent of N specimens and 70 per cent of R specimens have $J_{150} < 5 \times 10^{-4}$ emu g^{-1} while 85 per cent of A specimens have J_{150} below this value. (Anomalous (A) dykes are defined here as in Dagley *et al.* (i.e. with divergence angles of greater than 40° from the axial dipole field for the present geographic latitude of Mull).

More ideas concerning the meaning of the elongated group of southerly directed palaeomagnetic directions can be gained by comparing the Mull dyke result with results from other parts of the British Tertiary igneous province. Shallow reverse inclination, southerly declination units are found throughout the igneous province (Skye lavas and dykes (Khan 1960), Skye dykes (this paper), Ayrshire dykes (Raja 1965), Gt Cumbrae Island and Ayrshire dykes (Smith 1966), Shropshire and Staffordshire dykes (Dagley 1969) and Antrim lavas (Wilson 1970)).

This geographically widespread occurrence of shallow reverse directions of magnetizations can be explained in one of two possible ways: Igneous activity was short lived and contemporaneous over much of the igneous province and some atypical feature of the geomagnetic field, perhaps part of a polarity reversal, was widely recorded. Alternatively, the geomagnetic field spent rather longer in a non-centred axial dipole disposition during the period of igneous activity than during other geological periods. Igneous activity of different ages in different parts of the province then had a high probability of recording one of the periods of non-centred axial dipole behaviour. Currently much effort is being given to the reliable K/A dating of the basaltic rocks of this igneous province in order to choose between the different interpretations of the abundant precise palaeomagnetic information now becoming available.

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8. Summary and conclusions

In this investigation a previously deeply buried and thus considerably hydrothermally altered dyke swarm was investigated from the palaeomagnetic aspect. Reasons are given for believing that, notwithstanding the high degree of alteration, the a.f. cleaned NRM directions are at least in most cases very close to the original cooling TRM directions. However, the possibility may still exist that some form of directional remagnetization has taken place.

The palaeomagnetic results show a remarkable asymmetry with 39 closely grouped normal directions and a meridionally elongated group of 115 reverse and shallow reverse directions.

Possible explanations for this asymmetry are discussed. It seems likely that either igneous activity was short lived, and the field not sampled in a representative manner by the individual dykes, or that non-centred axial dipole behaviour of the field was more common during the Palaeogene than at other times.

J. M. Ade-Hall, P. Dagley, and R. L. Wilson
Sub-Department of Geophysics,
University of Liverpool,
Liverpool.

A. Riding:
Caldy Grange Grammer School,
West Kirkby, Cheshire.

A. Evans:
Department of Geodesy and Geophysics,
Madingley Rise,
Madingley Road,
Cambridge.

Present Address:
Department of Clinical Physics and Bioengineering,
Western Regional Hospital Board,
11 *West Graham Street,*
Glasgow C4.

P. J. Smith:
Department of Geology,
The Open University,
Walton Hall, Milton Keynes,
Bletchley, Buckinghamshire.

R. Skelhorne and T. Sloan:
Department of Geology,
Sir John Cass College,
Jewry Street,
London, E.C.

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